A compact-regime Einstein realization theorem on aspherical four-manifolds

23 September 2026. Research continuation; standard geometric compactness inputs are separated from the new application of the compact-core exclusion. The result concerns the original pointwise Ricci defect, not a replacement by a volume-normalized integral defect.

1. Statement and mathematical increment

For a closed Riemannian four-manifold define

$$\delta_-(g) = \| \operatorname{Ric}_g^\sharp + I \|_{L^\infty,\mathrm{op}}.$$

All Weyl norms below are tensor norms; a fixed equivalent curvature-operator norm changes only constants. An oriented manifold is **aspherical** when its universal cover is contractible.

Theorem A (compact-regime realization). Fix $D, v, \Lambda > 0$ and $2 < q < \infty$. There exists $\varepsilon(D, v, \Lambda, q) > 0$ with the following property. If a closed connected smooth oriented aspherical four-manifold M carries a smooth metric g satisfying

$$\operatorname{diam}(M, g) \leqslant D, \qquad \operatorname{Vol}(M, g) \geqslant v, \qquad \|W_g^+\|_{L^q(M,g)} \leqslant \Lambda, \qquad \delta_-(g) < \varepsilon, \tag{1}$$

then the **same smooth manifold** carries a smooth metric h with

$$\operatorname{Ric}_h = -h. \tag{2}$$

The threshold is uniform across the manifolds satisfying (1). One may replace W^+ by W^-, by reversing orientation. A pointwise bound on one Weyl half is a sufficient special case, since volume has a diameter-dependent upper bound under the Ricci bounds in (1).

No bound on $\nabla \operatorname{Ric}$, the full pointwise curvature, Euler characteristic, injectivity radius, or a separately supplied full curvature energy occurs in the theorem. The volume, diameter, and supercritical one-sided Weyl assumptions are genuine restrictions. In particular, the threshold is not dimension-only.

The preceding aspherical compact-core exclusion and exact no-bubbling theorem used Einstein regularity after rescaling a maximum of the full curvature. That proof does **not** apply verbatim to almost-Einstein metrics. The new argument normalizes a harmonic radius instead. Ricci convergence in the harmonic-coordinate elliptic equation supplies precisely the regularity needed for a smooth Einstein limit and for preservation of the smooth type; derivatives of the Ricci error are never estimated.

2. Inputs, with the regularity actually used

We use the following results.

1. The inherited compact-core lemma states that no large compact core of a nonflat smooth Ricci-flat anti-self-dual ALE four-manifold embeds smoothly into a closed aspherical four-manifold. Its proof treats both the negative-definite $b_2 > 0$ case and the rational-ball $b_2 = 0$ case. For the latter, lifting the core to the contractible universal cover and applying integral Mayer–Vietoris would give $\mathbb{Z}_{n^2} \cong \mathbb{Z}_n \oplus G$, $|G| = n$, which is impossible. The classification inputs are Wright and Suvaina.

2. Uniform harmonic-coordinate radii give pointed geometric $C^{1,\alpha}$ precompactness, with a loss of Hölder exponent before using the metric equation. On harmonic charts,

$$g^{ab}\partial_a\partial_b g_{ij} = -2\,\mathrm{Ric}_{ij} + Q_{ij}(g, \partial g), \tag{3}$$

where Q is quadratic in first derivatives. Bounded Ricci and controlled charts give local $W^{2,p}$ bounds for every finite p. Strong convergence of the right side gives strong $W^{2,p}$ convergence on smaller charts. These are the harmonic coordinate compactness tools of Anderson; the strong convergence step is also discussed explicitly by Zhang–Zhu.
3. Jiang–Naber's local L^2 curvature estimate applies to noncollapsed balls with two-sided bounded Ricci curvature, without any derivative bound on Ricci. With fixed diameter and positive total volume it gives a uniform global L^2 curvature bound in dimension four. Section 3 records the covering argument.
4. A complete Ricci-flat four-manifold with Euclidean lower volume growth and finite L^2 curvature is ALE, by the Bando–Kasue–Nakajima asymptotic theorem, Theorem 1.5; in dimension four the resulting ALE order is four. This result is applied only to a smooth, complete **exact Ricci-flat limit**, not to the original approximate metrics.

These inputs do not include an assertion that an almost-Einstein sequence converges smoothly. The convergence established below is $C^{1,\alpha}$ and strong $W^{2,p}$ for finite p, after diffeomorphisms. The limiting metric is smooth by (3).

3. Noncollapse and curvature energy from the bounded global regime

Fix a two-sided Ricci bound $|\mathrm{Ric}| \leqslant K$, diameter bound D, and volume lower bound v. Set $D_* = D+1$. Relative Bishop–Gromov comparison gives, for all $x \in M$ and $0 < r \leqslant D_*$,

$$\mathrm{Vol}\, B_r(x) \geqslant v\,\frac{V_{-K/3}(r)}{V_{-K/3}(D_*)} \geqslant c(K, D, v)r^4. \tag{4}$$

Here $V_{-K/3}$ denotes the four-dimensional model-ball volume; using a larger negative comparison constant causes no problem. The same comparison gives $\mathrm{Vol}(M) \leqslant V_{-K/3}(D_*)$.

Choose a fixed small radius $r_* = r_*(K, D) > 0$. A maximal disjoint collection of $r_*/4$-balls has uniformly bounded cardinality by (4) and the total-volume upper bound. The concentric $r_*/2$-balls cover M. On these balls the rescaled Ricci bound and volume lower bound meet Jiang–Naber's hypotheses, with constants depending only on K, D, v. Summing their estimates gives

$$\int_M |\mathrm{Rm}|^2 \leqslant E(K, D, v). \tag{5}$$

Thus neither a bound on $\chi(M)$ nor an additional curvature-energy assumption is needed. If one elects not to use the Jiang–Naber theorem, all subsequent statements remain valid with (5) supplied as an extra hypothesis and constants also depending on that energy bound.

4. Half-Weyl control prevents harmonic-radius collapse

Theorem B (harmonic compactness). Given $K, D, v, \Lambda > 0$, $q > 2$, and $0 < \alpha < 1$, there is a positive uniform lower bound for the $C^{1,\alpha}$ harmonic radius on every closed connected smooth oriented aspherical four-manifold with

$$|\mathrm{Ric}| \leqslant K, \quad \mathrm{diam} \leqslant D, \quad \mathrm{Vol} \geqslant v, \quad \|W^+\|_{L^q} \leqslant \Lambda. \tag{6}$$

Consequently this class is geometrically precompact in $C^{1,\beta}$, $\beta < 1$, and weak $W^{2,p}$, $p < \infty$, and has only finitely many diffeomorphism types.

Proof. Fix a standard harmonic-radius definition with metric comparability and scaled $C^{1,\alpha}$ norm bounded by fixed constants. Suppose its global minimum ρ_j, attained at x_j, tends to zero. Write

$$\widehat{g}_j = \rho_j^{-2} g_j, \quad r_h(\widehat{g}_j, x_j) = 1, \quad r_h(\widehat{g}_j, x) \geqslant 1 \quad \text{for all } x. \tag{7}$$

The rescaled Ricci tensors satisfy

$$|\operatorname{Ric}_{\widehat{g}_j}| \leqslant K\rho_j^2 \longrightarrow 0. \tag{8}$$

For any fixed R, (4) gives, for large j,

$$\operatorname{Vol}_{\widehat{g}_j} B_R(x_j) \geqslant c(K, D, v)R^4. \tag{9}$$

The lower volume and Ricci bounds ensure that the rescaled diameters tend to infinity. Indeed, the unscaled diameters have a positive lower bound from v and Bishop comparison.

The uniform harmonic charts in (7), bounded numbers of charts on each fixed-radius ball, and completeness produce pointed $C^{1,\beta}$ convergence on compact sets, $\beta < \alpha$, to a complete smoothable manifold (X, g_∞, x_∞). Initially the limiting metric has only the indicated low regularity. Equation (3) and (8) imply distributionally $\operatorname{Ric}(g_\infty) = 0$. Local elliptic bootstrapping in harmonic coordinates makes g_∞ smooth and Ricci-flat.

It is important that convergence can now be improved without differentiating (8). In compatible harmonic charts, subtract the limiting version of (3). For $u_j = g_j - g_\infty$, the right side of the resulting uniformly elliptic equation is a sum of

$$-2\operatorname{Ric}_{g_j}, \qquad Q(g_j, \partial g_j) - Q(g_\infty, \partial g_\infty), \qquad (g_\infty^{ab} - g_j^{ab})\partial_a\partial_b g_\infty. \tag{10}$$

Each tends to zero in L^p on compact smaller charts for every finite p, by (8), the initial C^1 convergence, and smoothness of the limit. Interior elliptic estimates imply

$$g_j \longrightarrow g_\infty \quad \text{strongly in } W^{2,p}_{\mathrm{loc}} \quad (p < \infty). \tag{11}$$

Choose $p > 4/(1-\alpha)$ to recover strong $C^{1,\alpha}$ convergence. Harmonic-coordinate transition maps are controlled by their own harmonic equations; the usual geometric compactness construction yields embeddings of compact subsets. More explicitly, take an open neighbourhood of any chosen smooth compact core of the limit and a finite harmonic atlas there. Controlled transition maps give a common C^1 differentiable structure and C^1 embeddings into the approximating smooth manifolds. A compatible smoothing of this atlas is unique up to diffeomorphism; C^1 approximation of each embedding by a smooth map remains an embedding on the compact core. This preserves its smooth type. No convergence of holomorphic curves or C^2 convergence of the approximating metrics is used.

The limit is not flat. If it were flat, (9) forces it to be Euclidean space: a complete flat quotient with an infinite deck group has sub-Euclidean volume growth, and a finite Euclidean isometry group acting freely is trivial. Equivalently, the ALE conclusion below also excludes every nontrivial smooth flat limit with (9). On a fixed Euclidean ball of radius greater than two there are harmonic charts with strict margins in all defining inequalities. Strong convergence in (11), followed by solving the harmonic coordinate Dirichlet problem on a slightly smaller ball, gives such a chart of radius greater than one on $(M_j, \widehat{g}_j, x_j)$. This contradicts (7). This is the harmonic-radius normalization argument, not an assertion that a pointwise maximum of curvature survives weak convergence.

Under the rescaling in (7), the half-Weyl norm has the decisive transformation rule

$$\int_{M_j} |W^+_{\widehat{g}_j}|^q \, d\mu_{\widehat{g}_j} = \rho_j^{2q-4} \int_{M_j} |W^+_{g_j}|^q \, d\mu_{g_j} \leqslant \rho_j^{2q-4} \Lambda^q \longrightarrow 0. \tag{12}$$

By (11), curvature converges strongly in local finite L^p spaces. The Hodge star also converges, so (12) implies $W^+_{g_\infty} = 0$. The full L^2 curvature is scale invariant in dimension four; (5), (11), and exhaustion imply

$$\int_X |\operatorname{Rm}_{g_\infty}|^2 \leqslant E(K, D, v). \tag{13}$$

Equations (9), (13), and Ricci-flatness place the complete nonflat limit within the Bando–Kasue–Nakajima ALE theorem. It is therefore a smooth nonflat ASD Ricci-flat ALE four-manifold. Its end is unique: a complete Ricci-flat manifold with two ends splits as a line times a compact manifold, which is incompatible with (9). The finite fundamental group, negative-definite intersection form, and the rational-ball alternatives used by the compact-core lemma are supplied by Wright's classification; simple connectivity is not assumed.

Choose a sufficiently large compact core $B \subset X$. The compact geometric convergence embeds B into M_j for large j, contradicting the inherited aspherical compact-core exclusion. Thus the assumed convergence $\rho_j \to 0$ is impossible, proving the uniform harmonic-radius bound. Standard finite-atlas geometric compactness, using the fixed diameter, now proves the final assertions of the theorem. □

Endpoint variation. The same argument permits the L^q bound in (6) to be replaced by uniform local equiintegrability of one Weyl half:

$$\sup_{(M,g)} \sup_{x \in M} \int_{B_r(x)} |W^+|^2 \, d\mu \leqslant \eta(r), \qquad \eta(r) \longrightarrow 0 \ (r \downarrow 0). \tag{14}$$

Indeed the integral on every fixed rescaled ball is bounded by $\eta(R\rho_j) \to 0$. Constants then depend on the common modulus η. A bare bounded critical L^2 norm does **not** supply (14); the proof makes no assertion at that unrestricted endpoint.

5. Proof of exact realization on the same smooth manifold

Suppose Theorem A fails. There are closed aspherical M_j and metrics satisfying its fixed D, v, Λ, q bounds such that $\delta_-(g_j) \to 0$, but no M_j admits a normalized negative Einstein metric. Discard finitely many terms so that $\delta_-(g_j) \leqslant 1/2$. This gives a fixed two-sided Ricci bound.

Theorem B and compact geometric convergence give, after a subsequence, a fixed closed smooth M_∞, diffeomorphisms $\phi_j : M_\infty \to M_j$, and

$$\phi_j^* g_j \longrightarrow g_\infty \quad \text{in } C^{1,\alpha} \tag{15}$$

for every fixed $\alpha < 1$, with weak local $W^{2,p}$ control. In these charts,

$$\operatorname{Ric}_{\phi_j^* g_j} = -\phi_j^* g_j + e_j, \qquad \|e_j\|_\infty \to 0. \tag{16}$$

Pass to the harmonic-coordinate equation (3). The strong convergence of the lower-order terms and (16) imply $\operatorname{Ric}_{g_\infty} = -g_\infty$ weakly. Elliptic bootstrapping makes the limit a smooth negative Einstein metric. The difference-equation argument (10) gives strong $W^{2,p}$ convergence for every finite p, if desired. Transporting g_∞ by ϕ_j gives a smooth metric with Ricci equal to minus itself on **each sufficiently late** M_j, a contradiction. This proves Theorem A. □

This argument establishes existence, not an effective Newton radius, uniqueness, or a canonical choice of the output. It also does not claim C^2, let alone smooth, convergence of the input metrics. Small, rapidly oscillating Ricci errors cannot be differentiated for free. Finite-p harmonic elliptic control is exactly what the topology-preserving limit argument needs.

6. The gauge-theoretic input can supply noncollapse

Corollary C. Fix $D, \Lambda, A_0 > 0$ and $q > 2$. In Theorem A, the explicit volume lower bound may be replaced by the following actual monopole supply: for every allowed input metric, there is an actual class a_g for which the scalar-curvature estimate holds and

$$(a_g^+)^2 \geqslant A_0. \tag{17}$$

The resulting threshold depends on D, Λ, q, A_0.

Proof. The actual monopole scalar estimate and $\delta_- \leqslant 1/2$ give

$$32\pi^2 A_0 \leqslant \int_M s_-^2 \, d\mu \leqslant 16(1+\delta_-)^2 \operatorname{Vol}(M,g).$$

Consequently

$$\operatorname{Vol}(M,g) \geqslant \frac{2\pi^2 A_0}{(1+\delta_-)^2} \geqslant \frac{8\pi^2 A_0}{9}. \tag{18}$$

Use this v in Theorem A. The class is allowed to depend on the metric. No assertion about an arbitrary formal cohomology vector is substituted for the actual scalar estimate. □

This supplies a genuine bridge between the established gauge budget and exact Einstein existence: the gauge input is used to prevent collapse, the one-sided curvature budget determines the possible blowup orientation, and asphericity excludes the resulting smooth ALE cores. It is not the false implication "no numerical obstruction, therefore existence." The independent geometric hypotheses remain.

7. Consequences for Yau26 and the precise remaining escape

Let M_j be closed oriented aspherical four-manifolds with no Einstein metric, and suppose $\delta_-(g_j) \to 0$. For every fixed $q > 2$, the following three properties cannot all hold uniformly:

$$\operatorname{diam}(g_j) \leqslant D, \qquad \operatorname{Vol}(g_j) \geqslant v > 0, \qquad \|W_{g_j}^+\|_{L^q} \leqslant \Lambda. \tag{19}$$

If the actual positive projection budget (17) is uniform, the volume condition is automatic. In that setting a hypothetical counterexample sequence must escape every joint diameter/half-Weyl bound. This is a statement about the approximate inputs themselves; unlike the earlier spectral escape theorem, it does not assume that the inputs are exact Einstein metrics or satisfy a rank-one Weyl spectral ansatz.

The theorem works equally for changing smooth types, because its compactness conclusion recovers the smooth type of each late term. It does not complete the negative Yau26 branch: arbitrary diameter, collapse without a gauge volume budget, and concentration of both Weyl halves remain outside (1). No assumption is silently made that the first-line scalar or spectral inequalities control those freedoms.

The meaningful increment is a theorem-level realization mechanism:

pointwise almost-Einstein input + bounded global regime + one-sided supercritical Weyl control
+aspherical topology $\Longrightarrow$ Einstein metric on the same smooth manifold.

The compactness, curvature-energy, and ALE classification theorems are not newly proved here. The continuation is their explicit composition with the inherited compact-core obstruction, including the missing almost-Einstein regularity argument. No worldwide-priority claim is made.

Comparison with an existing no-bubbling mechanism

Fine's 2017 twistor argument also excludes locally hyperkähler bubbles, but uses a different input: a curvature inequality making the twistor space tamed. His Corollary 4 gives compactness for normalized ASD Einstein metrics under diameter and volume bounds, without our asphericity assumption. His Remark 5 explicitly identifies the lack of C^2 convergence for bounded-Ricci sequences as an obstacle to using his holomorphic-curve persistence proof. We do not remove that obstacle for his theorem. Here no twistor taming or persistence is assumed; the compact-core contradiction is purely topological once the smooth ALE limit and core embedding have been obtained. Neither result subsumes the other with its hypotheses unchanged.

8. Primary sources and dependency trace

- M. T. Anderson, *Convergence and rigidity of manifolds under Ricci curvature bounds*, Invent. Math. **102** (1990), 429–445: https://doi.org/10.1007/BF01233434. Harmonic-radius and geometric compactness framework; equation (3) and the strong-convergence argument are specified above rather than hidden in an "Einstein compactness" citation.
- Q. S. Zhang and M. Zhu, *Bounds on harmonic radius and limits of manifolds with bounded Bakry–Émery Ricci curvature*, introduction and Section 2: https://arxiv.org/html/1705.10071. In particular their discussion distinguishes convergence in the classical Ricci case from the weaker Bakry–Émery setting and explains the strong Sobolev issue.
- W. Jiang and A. Naber, *L2 Curvature Bounds on Manifolds with Bounded Ricci Curvature*, Theorem 1.6: https://arxiv.org/html/1605.05583. This provides (5) under precisely a Ricci bound and local noncollapse, not Ricci derivative control.
- S. Bando, A. Kasue and H. Nakajima, *On a construction of coordinates at infinity on manifolds with fast curvature decay and maximal volume growth*, Invent. Math. **97** (1989), 313–349: https://doi.org/10.1007/BF01389045, Theorem 1.5. Its precise finite-energy Ricci-flat formulation is reproduced as Theorem 2.7 in R. Buzano and L. Yudowitz, *Bubble-tree convergence and local diffeomorphism finiteness for gradient Ricci shrinkers*: https://link.springer.com/article/10.1007/s00209-023-03272-z.
- E. P. Wright, *Quotients of gravitational instantons*: https://arxiv.org/pdf/1102.2442, and I. Suvaina, *ALE Ricci-flat Kähler metrics and deformations of quotient surface singularities*: https://arxiv.org/pdf/1102.2672. They are used in the inherited compact-core proof.
- For the actual scalar budget and its quantifiers, see the inherited all-sign stability proof and Bauer–Furuta class supply.
- J. Fine, *Limits of Riemannian 4-manifolds and the symplectic geometry of their twistor spaces*, Theorem 3, Corollary 4 and Remark 5: https://arxiv.org/html/1602.03829. This is a comparison, not an input to the proof.